

IPv6 Active Address Discovery in Few-Seed BGP Prefixes

Wenjian Zhang^{1b}, Guanglei Song^{1b}, Hanzhe Li^{1b}, Hai Lin^{1b}, Zihou Zhao^{1b}, Zhikang Mao^{1b}, Lin He^{1b}, *Member, IEEE*, Zhiliang Wang^{1b}, *Member, IEEE*, and Jiahai Yang^{1b}, *Senior Member, IEEE*,

Abstract—Accurate Internet-wide IPv6 active address discovery is essential for network measurement, security analysis, and asset census. However, the enormous IPv6 address space makes brute-force scanning infeasible. Existing target generation algorithms improve probing efficiency by learning seed addresses, but their performance degrades sharply in few-seed BGP prefixes, where only a limited number of seed addresses are available. In such cases, sparse distributions of seed addresses obscure structural regularities, while severe sampling bias can mislead probing away from dense active regions. We present 6Vision, a fine-grained IPv6 active address discovery method designed for few-seed BGP prefixes. 6Vision clusters structurally similar addresses, constructs augmented features, and learns dependencies both within and across addresses through image representations of seed addresses. This design enables the model to capture subtle address configuration regularities that are difficult to extract using conventional coarse-grained pattern-based methods. To further mitigate sampling bias, 6Vision introduces a reinforcement-learning-based online feedback mechanism that continuously updates the probing policy using online probing rewards and progressively guides probing toward denser active regions. Experiments on real-world demonstrate that 6Vision consistently outperforms state-of-the-art methods in few-seed BGP prefixes, improving hit rate by 45%–1,129% and covering 7%–278% more prefixes. Moreover, 6Vision can assist existing methods in probing few-seed BGP prefixes, boosting their effectiveness by 68%–330%. A six-month large-scale deployment further discovers 33.08M non-aliased active IPv6 addresses in few-seed BGP prefixes, substantially enriching existing IPv6 hitlists in previously underexplored regions and providing a valuable data for future research on IPv6 measurement and security analysis.

Index Terms—IPv6, few-seed BGP prefixes, active address detection, target address generation, network measurement.

I. INTRODUCTION

GLOBAL active address scanning is a fundamental technique for large-scale Internet measurement and security analysis. It supports network topology inference by revealing interconnections among Internet nodes and networks [1], [2], enables service discovery, upper-layer fingerprinting, and resource census for understanding Internet service deployment [3]–[7]. It also facilitates attack-surface analysis as well

as the detection of exposed or vulnerable critical infrastructure [8]–[12]. With the exhaustion of the IPv4 address space, IPv6 deployment has accelerated worldwide. For example, as of February 2026, more than 46% of users accessed Google services over IPv6 [13]. Although high-speed scanners such as ZMap [14] and Masscan [15] can exhaustively scan the entire IPv4 address space in about 45 minutes, extending this brute-force strategy to IPv6 is infeasible, as the enormous size of the IPv6 address space would require millions of years.

To address the impracticality of brute-force scanning in the IPv6 address space, researchers have developed efficient target generation algorithms (TGAs), such as AddrMiner-S [16] and 6Scan [17], to discover active addresses. The key idea of TGAs is to learn address configuration, such as patterns¹ from seed addresses, i.e., known active addresses, and then use the learned configuration to predict and probe target addresses, i.e., other potentially active addresses. Although existing large-scale TGAs have achieved high hit rates, e.g., AddrMiner-S and 6Scan both report hit rates exceeding 30%, the active addresses they discover are often heavily concentrated in a small number of BGP prefixes [16], [17]. We attribute this concentration to the abundance of seed addresses in these prefixes, which makes address configuration patterns easier to learn. Consequently, existing TGAs tend to allocate probing resources to such BGP prefixes. In contrast, BGP prefixes with only a few seed addresses, which we refer to as few-seed BGP prefixes, often suffer from sparse distributions and severe sampling bias, which substantially limits the ability of existing methods to uncover active addresses. More importantly, such BGP prefixes account for 66.78% of the prefix space (see §III-B2), indicating that a large portion of the IPv6 space remains insufficiently explored. A representative example illustrates this issue. The BGP prefix 2a01:e0a::/38 contains approximately 0.5M seed addresses. With a probing budget² of 1M, AddrMiner-S achieves a hit rate of 64% in this prefix. By contrast, the BGP prefix 2001:250:307c::/48 contains only three sparsely distributed seed addresses. Under the same probing budget of 1M, AddrMiner-S discovers only five active addresses.

As in Figure 1, we find that this performance degradation is attributed to two fundamental issues arising from the extremely limited number of seed addresses in these prefixes.

- **Sparse distribution.** Seed addresses within these prefixes are often sparsely distributed and exhibit substantial heterogeneity in their assignment patterns, render-

Wenjian Zhang, Hai Lin, Lin He, Zhiliang Wang, Jiahai Yang are with the Institute of Network Science and Cyberspace, Tsinghua University, Beijing 100084, China and also with Zhongguancun Laboratory, Beijing 100094, China (E-mails: {zhang-wj23, linhai22}@mails.tsinghua.edu.cn, {he-lin}@tsinghua.edu.cn, {wzl, yang}@cernet.edu.cn).

Guanglei Song is with Zhongguancun Laboratory, Beijing 100094, China (E-mail: songgl@mail.zgclab.edu.cn).

Hanzhe Li is with College of Artificial Intelligence, Xi'an Jiaotong University, Xi'an 710049, China (E-mail: lhz2023@stu.xjtu.edu.cn).

Zihou Zhao is with Minzu University of China, Beijing 100081, China (E-mail: 23160169@muc.edu.cn).

Zhikang Mao is with the School of Computer and Cyber Sciences, Communication University of China, Beijing 100024, China (e-mail: tab-tac@cuc.edu.cn).

¹Practices for configuring addresses, e.g., 2001:0da8:****:1, where * denotes a wildcard representing a hexadecimal digit (from 0x0 to 0xf) [18].

²The number of generated target addresses.

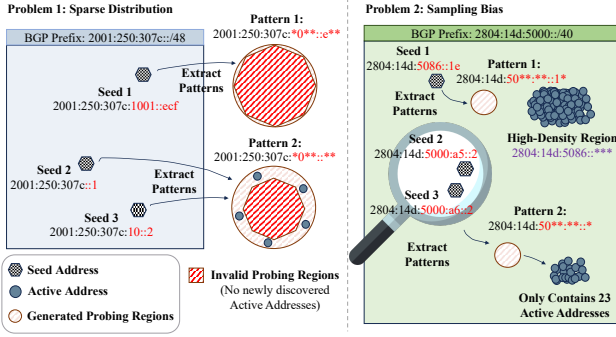


Fig. 1. Two issues in few-seed BGP prefixes during address discovery.

ing coarse-grained pattern extraction unreliable. For example, under the BGP prefix 2001:250:307c::/48, only three seed addresses are available, and they differ significantly in their configuration regularities. Existing methods [16], [17] thus derive coarse patterns such as 2001:250:307c:*0*::* and 2001:250:307c:*0*::* with 1M budget. These patterns have an enormous address space but contain only five active addresses. This suggests that coarse-grained patterns, when inferred from sparse and heterogeneous seed addresses, tend to overgeneralize the address space and thus direct probing toward largely invalid regions.

- **Sampling bias.** With only a few seed addresses, the observed samples may poorly represent the actual active address space, causing existing methods to overlook high-density regions. For example, under the BGP prefix 2804:14d:5000::/40, three seed addresses are observed, from which an existing method extracts the patterns 2804:14d:50*::* and 2804:14d:50*::*. Yet these patterns contain only 23 active addresses. In contrast, although 2804:14d:5086::1e appears structurally different from the other two seeds, we find that the dense active region is actually concentrated in the pattern 2804:14d:5086::*, which contains 536 active addresses. This example highlights the inability of existing methods to correct probing directions under biased and insufficient samples, thereby missing dense and high-yield target regions.

To address the limited effectiveness of existing TGAs in few-seed BGP prefixes, we first identify such prefixes based on prior work [19] and Internet-wide measurements. Then, we design 6Vision to specifically address this problem. The key intuition of 6Vision is that, although seed addresses are sparse and biased in the few-seed BGP prefixes, their underlying configuration regularities remain exploitable. To capture these regularities, 6Vision transforms seed addresses into image representations, enabling fine-grained learning of both intra-address and inter-address correlations, and continuously refines its probing strategy using online feedback. Specifically, to expand the seed addresses from all few-seed BGP prefixes, we first aggregate the seed addresses from these prefixes to form the input seed address set. 6Vision then maps this set into a learned latent space and groups addresses with

similar configurations. It then constructs augmented features by concatenating multiple addresses from the same group into image representations. This design not only enlarges the effective training set under limited seeds, but also exposes richer contexts for learning address configurations. Then 6Vision employs Gated PixelCNN to jointly model intra-address and inter-address dependencies, thereby learning fine-grained distributional regularities that are difficult to recover from sparse seeds with coarse-grained generation strategies. Because of the sampling bias, an accurate offline model alone is insufficient for efficient probing. 6Vision therefore further introduces a reinforcement learning based feedback mechanism that uses online probing feedback to continuously adjust the probing direction. This allows the model to move beyond the limited perspective of the initial seeds and gradually steer probing toward more promising high-density regions. Overall, 6Vision provides an effective solution for IPv6 active address discovery under the few-seed BGP prefixes. Beyond directly improving discovery performance in few-seed BGP prefixes, 6Vision enlarges the pool of learnable addresses, thereby enabling existing methods to probe such prefixes more effectively. In this sense, 6Vision functions not only as a standalone discovery approach but also as a practical means of broadening the applicability of existing TGAs. Our work is available at <https://github.com/zwijsndy/6Visionpro>.

The contributions of this paper are as follows:

- We identify the few-seed problem in IPv6 active address discovery and show that it is prevalent in the real world. Our analysis indicates that, in the few-seed BGP prefixes, limited seed addresses fail to provide enough information for existing TGAs to generate high-quality targets.
- A fine-grained image-based target generation method is proposed specifically for the few-seed BGP prefixes. By clustering structurally similar addresses, constructing augmented address images, and learning dependencies both within and across addresses, 6Vision captures subtle structural regularities that are easily obscured by the sparse distribution of active addresses.
- An online feedback optimization mechanism based on reinforcement learning is further designed. By updating the probing policy with online probing rewards, while stabilizing training with entropy regularization and a KL constraint, 6Vision progressively refines probing toward denser active regions and mitigates sampling bias.
- Real-world evaluations show that 6Vision improves the accuracy by 45%–1,129% over state-of-the-art methods and boosts the effectiveness of existing methods by 68%–330%. Moreover, a six-month deployment discovers 33.08M non-aliased active addresses, substantially enriching IPv6 hitlists in previously underexplored regions.

This paper extends our conference paper [20]. Compared with the conference version, this journal version provides a more comprehensive analysis of the few-seed problem, enhances 6Vision with improved structural modeling and reinforcement-learning-based online feedback optimization, substantially expands the experimental evaluation with parameter studies and interpretability analysis, and reports a six-

month large-scale deployment that discovers a larger amount of non-aliased active IPv6 addresses in few-seed BGP prefixes.

II. RELATED WORK

Prior studies on IPv6 active address discovery have primarily focused on designing target generation algorithms (TGAs) that take seed addresses as input and generate target addresses for probing. The central assumption behind these methods is that seed addresses contain useful information about the underlying IPv6 addressing scheme, and that such information can be exploited to reveal additional active addresses in the same address space. Under this paradigm, seed addresses play a critical role in determining the effectiveness of subsequent probing. Existing TGAs can be broadly divided into two categories: statistical methods [16]–[19], [21]–[31] and machine-learning-based methods [9], [32]–[35]. Statistical methods mainly rely on explicit analysis of density, structure, or hierarchy in seed addresses, whereas machine-learning-based methods attempt to learn latent address patterns through data-driven generative models.

A. Statistical methods

Statistical methods can be further divided into three categories: density-based methods [16], [19], [21], [23], [36], structure-based methods [18], [19], [22], and hierarchy-based methods [17], [19], [24], [25], [27]–[31].

1) *Density-based methods*: Density-based methods aim to generate target addresses by learning the density characteristics of seed addresses. Their core intuition is that dense seed regions are more likely to correspond to active regions in the real IPv6 address space, and thus deserve more probing effort. Ullrich et al. [21] proposed an automatic pattern generation algorithm, which marked the beginning of this line of research. 6Gen [23] adopts a bottom-up agglomerative hierarchical clustering (AHC) strategy to group seed addresses into dense regions, thereby enabling pattern extraction and fine-grained address generation. DET [26] uses entropy as the segmentation metric and employs a top-down divisive hierarchical clustering (DHC) strategy to partition the density space tree according to minimum entropy. Building on this idea, AddrMiner-S [16], [19] further improves generation efficiency by incorporating a dynamic feedback mechanism based on Thompson sampling [37]. These methods have shown strong practical performance when dense seed regions are available [38]. However, their effectiveness is closely tied to the availability and representativeness of seed addresses. In the few-seed BGP prefixes, dense regions are often difficult to identify reliably, and sampling bias in the seeds can easily mislead the generation process.

2) *Structure-based methods*: Structure-based methods analyze the internal organization and configuration patterns of seed addresses and generate target addresses by modeling dependencies among address components. Entropy/IP [22] probabilistically models IPv6 addresses using information-theoretic techniques, with the goal of capturing structural regularities in address assignment. Subsequent work further refined this idea through entropy-based clustering to reveal more

advanced structural patterns [18]. Compared with density-based methods, structure-based methods are better aligned with the observation that IPv6 addresses often follow non-random configuration rules. However, because interface identifiers (IIDs) can be generated through diverse mechanisms, such as EUI-64, randomized identifiers, embedded values, or protocol-specific formats, structural regularities are often heterogeneous and difficult to model accurately.

3) *Hierarchy-based methods*: Hierarchy-based methods partition seed addresses into subnet spaces according to hierarchical characteristics and then generate targets within the resulting address regions. 6Tree [24] treats IPv6 addresses as high-dimensional vectors and uses top-down DHC to build a hierarchical space tree in linear time. 6Hit [25] incorporates reinforcement learning to allocate exploration resources according to regional scanning rewards. 6Scan [17] further improves 6Tree by introducing region identifier encoding, which allows the search direction to be adjusted efficiently during scanning. HMap [31] extends this line of work by combining DHC and AHC for better space-tree construction. 6Graph [27] applies graph-theoretic pattern mining to generate address patterns, while AddrMiner-F [19] extracts and selects patterns through graph clustering and similarity matching. 6Forest [28] uses an enhanced isolation forest to remove outlier seed addresses, and 6Loda [30] further refines the space-tree partitioning strategy. Hierarchy-based methods are effective for organizing large address spaces and guiding exploration toward promising regions, but they remain highly dependent on seed quality. In the few-seed BGP prefixes, sparse distribution and sampling bias of seed addresses may lead to hierarchical structures that do not accurately capture the true active regions of the network.

B. Machine-learning-based methods

Machine-learning-based methods generally treat seed addresses as sequences and formulate target generation as a generative modeling task. Their key idea is to learn latent representations of seed addresses and then produce new target addresses that follow similar regularities. Compared with statistical methods, these approaches are expected to capture more complex dependencies and implicit configuration regularities.

6GAN first applies entropy clustering [18] for fine-grained classification of seed addresses, and then generates targets using a generative adversarial network (GAN) [39] combined with reinforcement learning. 6Sense [9] employs a long short-term memory (LSTM) [40] network to learn the /64 prefix patterns of addresses. More recently, 6PTG [35] uses a guided diffusion model to learn the distribution of active addresses. Despite their flexibility, existing machine-learning-based methods still face three major limitations in IPv6 probing. First, in the few-seed BGP prefixes, the small number of seeds results in insufficient learnable features, making fine-grained pattern learning difficult. Second, they often treat addresses independently or as simple sequences, without fully capturing both intra-address dependencies and inter-address correlations. Third, they remain sensitive to sampling bias in the seed set,

TABLE I
STATISTICS OF IPV6 ACTIVE ADDRESS DATASETS

Dataset	Addresses	Active	BGP Prefixes	ASes
Scamper	8.95M	1.54M	41,988	14,308
RIPE Atlas	0.61M	0.27M	55,915	17,030
CAIDA DNS	1.02M	6.71M	65,318	17,996
Gasser	19.16M	13.44M	62,655	19,653
Total	29.74M	15.24M	87,347	23,135

which can distort the learned distribution and lead to inefficient probing.

In conclusion, although statistical and machine-learning-based methods can achieve high hit rates when sufficient seed addresses are available, their accuracy degrades severely in the few-seed BGP prefixes due to the lack of sufficient active addresses for learning. Therefore, achieving high-accuracy probing in the few-seed BGP prefixes and enriching seed addresses so that existing methods can still perform effectively have become urgent challenges.

III. BACKGROUND AND MOTIVATION

A. Background

An IPv6 address is 128 bits long and typically represented as eight groups of four hexadecimal digits separated by colons. A unicast IPv6 address consists of three components: the global routing prefix, the subnet identifier, and the Interface Identifier (IID). IPv6 addresses exhibit the following key characteristics: (1) Vast Address Space: The IPv6 address space comprises 2^{128} addresses, which is vastly larger than the 2^{32} space of IPv4, thereby alleviating IPv4 address exhaustion. (2) Diverse IID Allocation Methods: The IID can be generated using various schemes, including random [41], [42], low-byte [41], IPv4 embedding [43], Teredo tunneling [44], EUI-64 embedding [45], modified EUI-64 [45], wordy [41], and ports [46]. (3) Sparse Active Address Space: Despite the enormous size of the IPv6 address space, the number of active network devices is relatively limited. Consequently, IPv6 active addresses are extremely sparsely distributed across the overall address space.

B. Motivation

1) *Prevalence of the Few-Seed Problem:* Consistent with prior work [38], we collected the latest publicly available IPv6 address datasets in March 2026 from three major sources: (1) addresses obtained by resolving domain names through AAAA record queries [47], (2) router topology datasets derived from traceroute measurements, including RIPE Atlas [48] and Scamper [49], and (3) the Gasser hitlist [50]. To characterize their network coverage, we further downloaded 246,183 announced IPv6 BGP prefixes from pyasn [51], whose database is provided by the Route Views Project [52]. For each dataset, we measured the number of addresses, the number of currently active addresses, and the number of covered BGP prefixes and ASes, as summarized in Table I. In total, the aggregated hitlist contains 29.74M addresses, among which 15.24M are active. These active addresses span 87,347 BGP prefixes and 23,135 ASes. However, their distribution across BGP prefixes is highly uneven. Figure 2 shows the CDF of IPv6 active

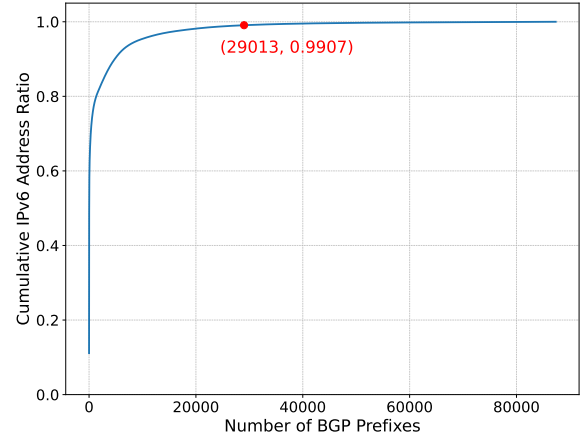


Fig. 2. CDF of IPv6 active addresses distributed across BGP prefixes.

addresses over BGP prefixes. A substantial proportion of BGP prefixes contain only a limited number of seed addresses; specifically, 66.78% of BGP prefixes have fewer than 10 seed addresses. This corresponds to the spanning from the 29,013th (red points in the figure) to the 87,347th BGP prefix. Therefore, BGP prefixes with only a few seed addresses are prevalent in the IPv6 hitlist. This prevalence highlights the severity of the few-seed problem.

2) *Definition of the Few-Seed Problem:* Existing TGAs often struggle when only a limited number of seed addresses are available. To characterize this challenge, we first identify the BGP prefixes that give rise to such conditions. Following prior work [19], we define a BGP prefix as a few-seed BGP prefix if it contains no more than a given threshold number of seed addresses. To evaluate the overall effect of seed scarcity on TGA performance, rather than the idiosyncrasies of individual prefixes, we aggregate seed addresses from all few-seed BGP prefixes, and use each resulting seed set as the input of TGAs. For each threshold, we first count the number of seed addresses, non-aliased seed addresses, and BGP prefixes containing non-aliased seed addresses, together with their proportions, as reported in Table II. Since aliased prefixes³ [18], [38] and aliased addresses⁴ will waste probing resources [18], [38], we apply APD [18] to strictly remove them and use only non-aliased addresses as seed addresses in the following experiments. We then evaluate seven representative TGAs that support large-scale probing under a target budget of 10M addresses. Then, we compare their hit rates under different few-seed thresholds. As shown in Figure 3, we observe that the hit rates of all TGAs generally increase with the threshold and then gradually level off, as more seed addresses provide richer information for learning address patterns. When the threshold is 1, all methods achieve the lowest hit rates, close to zero. Increasing the threshold to 10 improves the performance of most methods, but the overall hit rates remain low, with all methods staying below 10%. In contrast, raising the threshold

³Vast address ranges may map to the same physical device, a phenomenon known as IPv6 aliasing. A prefix is considered aliased if the entire prefix is responsive and maps to a single device or a small set of devices.

⁴Addresses within an aliased prefix are referred to as aliased addresses.

TABLE II
STATISTICS UNDER DIFFERENT FEW-SEED THRESHOLDS

Threshold	1	10	100	250	500	750	1000
Addr.	0.03M	0.15M	0.81M	1.30M	2.59M	3.14M	3.35M
Non-alias Addr.	0.03M	0.15M	0.69M	1.10M	1.50M	1.79M	1.99M
BGP Prefixes	30,427	58,334	75,349	77,904	79,117	79,591	79,829
BGP Ratio (%)	34.83	66.78	86.26	89.19	90.58	91.12	91.39

Addr.: addresses; Non-alias Addr.: non-aliased addresses.

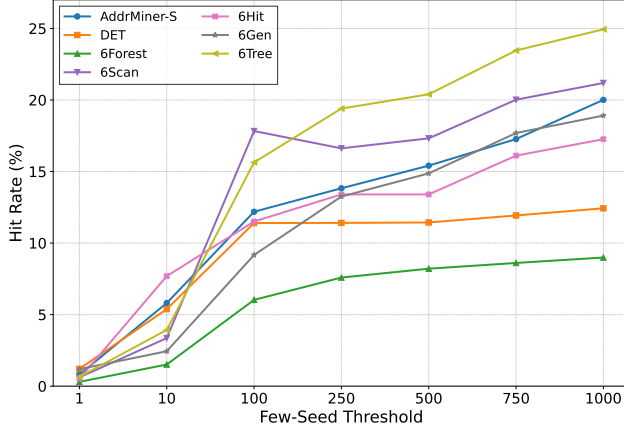


Fig. 3. Hit rates of different TGAs under different few-seed thresholds.

to 100 leads to a much more substantial improvement across all methods. Beyond this point, the gains become marginal and the performance tends to stabilize. Therefore, we set the threshold for few-seed BGP prefixes to 10. Under this threshold, 58,334 BGP prefixes are classified as few-seed BGP prefixes, accounting for 66.78% of all prefixes, and exert a pronounced negative impact on the hit rates of existing TGAs.

3) *Solution Approach*: The above results show that few-seed BGP prefixes are widespread in IPv6 hitlists and remain a major challenge for existing TGAs. Because only a few seed addresses are available in these prefixes, the information they provide is often insufficient for TGAs to capture meaningful addressing patterns, leading to low-quality target generation and poor probing accuracy. To address this limitation, our core idea is to enrich seed addresses in the few-seed BGP prefixes before applying existing TGAs. Specifically, we need to design an address enrichment method tailored to the few-seed BGP prefixes to effectively enrich the available seed addresses. By providing TGAs with more informative and enriched addresses, this enrichment step improves the effectiveness of their pattern extraction and target generation.

IV. DESIGN OF 6VISION

A. Overview

IPv6 target generation in the few-seed BGP prefixes is hindered by sparse seed distribution and sampling bias. To address these issues, we propose 6Vision, whose core idea is to transform seed addresses into image representations, enabling the model to capture subtle structural regularities both within and across addresses, and to further refine the probing direction using online feedback. To expand seed addresses across all few-seed BGP prefixes, we first combine the seed addresses from these prefixes into a single input seed set and then use

6Vision to probe the corresponding address space. As shown in Figure 4, 6Vision consists of four components. First, it clusters seed addresses based on structural features, so that addresses with similar configurations are grouped together. Second, it augments features by concatenating multiple addresses within each cluster into image representations, enriching the learnable patterns under limited seed addresses. Third, it employs Gated PixelCNN to jointly model intra-address and inter-address dependencies, enabling fine-grained distribution learning from the augmented features and alleviating the weak learning effectiveness caused by sparse seed distribution. Finally, it introduces a reinforcement learning based feedback mechanism to adapt the probing direction online and steer probing toward more promising high-density regions, thereby mitigating the sampling bias of the initial seed addresses.

B. Address clustering

Prior work [34] shows that fine-grained address clustering can improve the learning effectiveness of IPv6 address discovery models. We therefore cluster the seed addresses before model training. Each IPv6 address is encoded as a 128-bit binary vector to enable fine-grained modeling. Directly clustering in the raw 128-dimensional bit space is unreliable. As noted in [53], in high-dimensional spaces, the contrast between the nearest and farthest samples under Euclidean distance diminishes, degrading the usefulness of distance-based similarity. This issue is exacerbated in the few-seed BGP prefixes, where many bit positions exhibit weak regularity and high variability; clustering in the original space then introduces substantial noise and can mask meaningful structural patterns.

To address this, we first map addresses into a latent space and perform clustering on the learned representations. Deep learning models are effective for data with compositional and hierarchical structure [54], and IPv6 addresses exhibit such hierarchical characteristics [17], [22]. Consequently, low-dimensional latent vectors can preserve salient structure while reducing noise and redundancy. Since hierarchical clustering has also been shown to be effective for partitioning IPv6 address spaces [16], [17], [24], [26], we adopt it as our clustering algorithm.

Concretely, we employ a variational autoencoder (VAE) [55] to learn compact latent representations of seed addresses. The encoder maps each 128-bit vector into a lower-dimensional latent space, and the decoder reconstructs the original address from the latent code. In this way, the VAE extracts compact feature representations that capture the major structural regularities of the addresses. After training, the encoder outputs a latent feature vector for each seed address, and we apply hierarchical clustering to these latent features to partition the seed addresses into fine-grained clusters.

C. Feature augmentation

In the few-seed BGP prefixes, seed addresses are typically sparsely distributed over the target address space. As a result, existing methods, such as 6Scan and AddrMiner-S, tend to extract only coarse-grained patterns and thus struggle to capture the underlying address configuration regularities from

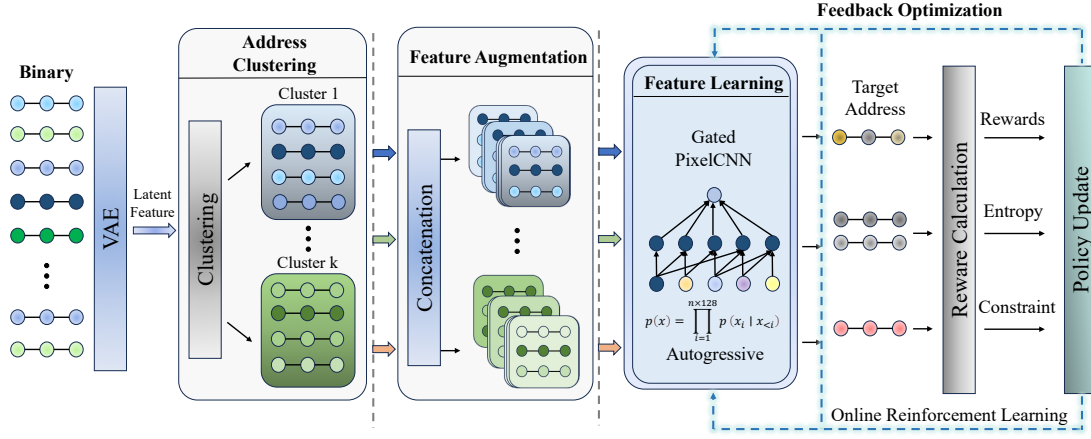


Fig. 4. System overview of 6Vision. It consists of four modules: address clustering, feature augmentation, feature learning, and feedback optimization.

limited seeds. To alleviate this limitation, we employ a deep learning model to learn fine-grained configuration patterns and enable more accurate address prediction. However, the effectiveness of deep models critically depends on the quality and richness of the learnable features. We therefore perform feature augmentation before model training.

Inspired by prior studies on representation learning [56], we augment features by concatenating similar addresses within each cluster. The intuition is that addresses grouped into the same cluster share similar configuration characteristics; aggregating them can expose more stable local structures and enrich the patterns available for learning. Specifically, for each cluster obtained in the previous stage, suppose that it contains N seed addresses. Since these addresses exhibit similar configuration patterns, we randomly sample n addresses from the cluster and concatenate them row by row. In this way, n individual 1×128 binary vectors are transformed into a $n \times 128$ image representations. Under this construction, a cluster with N addresses can generate at most $\binom{N}{n}$ distinct concatenated features. This augmentation strategy substantially increases the number of learnable training samples, thereby improving the ability of the downstream deep model to learn robust configuration patterns from limited seeds. More importantly, by jointly organizing addresses with similar structural characteristics into a unified image representation, the model is better positioned to capture complex and fine-grained dependencies that are difficult to identify from individual addresses alone.

D. Feature learning

After feature augmentation, each cluster yields image representations that aggregate multiple addresses with similar configurations. Compared with individual addresses, these features expose richer structural regularities and provide a more informative basis for model learning. The next step is therefore to learn the distribution of these augmented features and exploit it for accurate address prediction.

A key observation is that the target prediction task involves two types of dependencies. The first is *intra-row dependency*: within a single IPv6 address, the value of each bit is often constrained by preceding bits, reflecting the hierarchical structure of address configuration. The second is *inter-row*

dependency: addresses within the same cluster share similar configuration semantics, so adjacent rows in an augmented feature can provide valuable contextual cues for predicting one another. Existing coarse-grained methods [9], [34] do not explicitly model these two forms of dependency, which limits their effectiveness in the few-seed BGP prefixes. To jointly model both, we adopt Gated PixelCNN, an autoregressive convolutional generative model with gating [57], [58]. Given an augmented feature $x \in \mathbb{R}^{n \times 128}$, it factorizes

$$p(x) = \prod_{i=1}^{n \times 128} p(x_i | x_{<i}), \quad (1)$$

where x_i is the i -th element in row-major order, and $x_{<i}$ denotes all preceding elements.

Under our encoding mechanism, each training instance is formed by randomly sampling n seed addresses from the same cluster, binarizing them, and concatenating them row-wise into an $n \times 128$ representation. The model is then trained autoregressively to estimate the joint distribution of all bits in row-major order. In other words, each bit is predicted conditioned on all previously visible bits in the valid receptive field, including earlier bit positions in the same row and the rows above. This setup explicitly encourages the model to exploit both within-address sequential regularities and cross-address shared patterns. Autoregressive validity is enforced via masked convolutions (vertical and horizontal masks), ensuring that each prediction depends only on prior context, while stacking multiple layers progressively expands the receptive field to cover the required upper-left dependency region.

E. Feedback optimization

In the few-seed BGP prefixes, seed addresses often suffer from severe sampling bias. As a result, existing feedback methods [16], [25] become unreliable: limited seed addresses provide insufficient evidence to identify potential high-density regions, resulting in weak feedback and inefficient allocation of subsequent probing budgets. To address this issue, we introduce a reinforcement learning mechanism that formulates probing as a sequential decision process and adapts the probing policy online based on probing feedback. In this way, the

policy can progressively discover and focus on high-density regions even under limited seed addresses, while preserving address diversity and preventing excessive drift from the distribution learned by the initial model.

In our reinforcement learning framework, the address generation model serves as the policy and the real network as the environment. The state s denotes the partially generated bit sequence, and the action a corresponds to the next-bit generation decision. After probing, the environment returns a reward r based on the observed response. We assign positive rewards only to non-aliased active addresses within the few-seed BGP prefixes, while assigning negative rewards to inactive, aliased, and out-of-prefix addresses, thereby constraining the generated active addresses to remain in the few-seed BGP prefixes.

Inspired by GRPO [59], we maintain a dynamic reward baseline b_t to stabilize training, where b_t tracks the average feedback via an exponential moving average:

$$b_t = (1 - \alpha)b_{t-1} + \alpha\bar{r}_t, \quad (2)$$

where $\bar{r}_t$ is the mean reward of the t -th batch in the round and $\alpha \in (0, 1]$ is the smoothing coefficient. We then define the advantage as

$$A_t = r_t - b_t, \quad (3)$$

where r_t is the reward of the current sample. This formulation reduces reward variance and yields a more stable signal for policy updates. On top of this formulation, we optimize the generation policy using a policy-gradient objective. Let $\pi_\theta(a | s)$ denote the probability that the current policy generates action a under state s . The policy-gradient loss is defined as

$$\mathcal{L}_{pg} = -\mathbb{E}[\log \pi_\theta(a | s) \cdot A_t]. \quad (4)$$

Here, $\log \pi_\theta(a | s)$ denotes the log-probability of generating the current address. In our binary autoregressive address generation, this probability is obtained by summing the conditional log-probabilities of all address bits. Consequently, generated samples whose rewards exceed the baseline are reinforced, while those with below-baseline rewards are suppressed. This policy-gradient term therefore drives the model toward regions with higher expected utility and constitutes the core of the feedback optimization procedure.

However, optimizing solely with reward signals can introduce two major issues. First, the policy may gradually collapse to a small set of high-reward regions, reducing generation diversity. Second, it may drift too far from the distribution before feedback, thereby destroying useful structural regularities. To address these problems, we further incorporate an entropy regularization term and a KL-divergence constraint into the objective.

The entropy regularization term is used to preserve policy stochasticity and maintain exploration capability. For binary address generation, the entropy can be written as

$$\mathcal{H} = -\sum_{i=0}^{127} [p_i \log p_i + (1 - p_i) \log(1 - p_i)]. \quad (5)$$

where p_i is the predicted probability that the i -th bit takes value 1. This term discourages the policy from becoming

overly deterministic in a few known and sufficiently explored regions, thus helping preserve generation diversity.

To prevent the updated policy from deviating excessively from the initial distribution before feedback, which could lead to unstable and degraded generation quality, we further impose a KL-divergence constraint. Let π_{ref} denote the frozen trained generator, which corresponds to the initial model before feedback. The KL term is defined as

$$D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) = \sum_{i=0}^{127} \left[p_i^{(\theta)} \log \frac{p_i^{(\theta)}}{p_i^{(\text{ref})}} + \left(1 - p_i^{(\theta)}\right) \log \frac{1 - p_i^{(\theta)}}{1 - p_i^{(\text{ref})}} \right]. \quad (6)$$

where $p_i^{(\theta)}$ and $p_i^{(\text{ref})}$ denote the predicted probabilities for the i -th bit under the current policy and the reference policy before feedback, respectively. This constraint helps preserve the statistical regularities captured during initial training and reduces the risk of excessive policy drift or catastrophic forgetting under sparse and noisy feedback. Combining the above components, the final objective is

$$\mathcal{L} = \mathcal{L}_{pg} - \lambda_H \mathcal{H} + \lambda_{KL} D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}), \quad (7)$$

where λ_H and λ_{KL} control the strengths of entropy regularization and KL constraint, respectively. Together, these terms allow the model to exploit probing feedback for higher hit rates while maintaining sufficient exploration and distributional stability. Moreover, because active IPv6 addresses are extremely sparse, raw feedback would result in policy updates dominated by negative samples and would weaken positively rewarded samples during training, increasing their proportion in each batch and improving optimization efficiency and stability.

V. EXPERIMENTAL EVALUATION

A. Experimental Setup

1) *Experimental data*: To enable all TGAs to expand the seed set in as many few-seed BGP prefixes as possible, we aggregate seed addresses from the 58,334 few-seed BGP prefixes discussed in §III-B2, yielding a total of 148,135 non-aliased seed addresses. We use ZMapv6 [14] to perform ICMPv6 scans, as prior work has shown that active IPv6 addresses are most likely to respond to ICMPv6 probes [26]. We send one probe to each address and define responsive addresses as active. In addition, because the presence of aliased prefixes can lead to substantial waste of probing resources, we use APD [18] to identify and remove aliased addresses.

2) *Evaluation Metrics*: We evaluate the performance of TGAs using the commonly adopted accuracy metric, *hit rate*. Since our primary goal is to address the limitations of existing TGAs in few-seed BGP prefixes, we further consider three additional metrics tailored to this setting.

- **Hit rate**. Consistent with prior work [38], *hit rate* measures the fraction of generated target addresses that are active, excluding addresses already contained in the seed addresses

and aliased addresses. Formally, given a set of seed addresses S , generated target addresses C , active addresses T , and aliased addresses A detected from C , the hit rate is defined as

$$\text{hit rate} = \frac{|((C \cap T) \setminus (C \cap A)) \setminus S|}{|C|}. \quad (8)$$

- **Prefix hit rate.** To evaluate the ability of TGAs to enrich active addresses within the few-seed BGP prefixes, we define *prefix hit rate*. It is defined similarly to *hit rate*, but restricts the evaluation to the few-seed BGP prefixes. Specifically, we consider only generated target addresses that fall within the few-seed BGP prefixes, and we count an address as a hit only if it is active and non-aliased under this restriction; addresses already contained in the seed addresses remain excluded.
- **Coverage Ratio (CR).** *Coverage Ratio (CR)* measures the fraction of few-seed BGP prefixes covered by the active target addresses, i.e., the fraction of prefixes in which at least one active target address is discovered. Let P denote the set of few-seed BGP prefixes, and let a_p denote the number of active target addresses discovered in prefix p . Then, *CR* is defined as

$$CR = \frac{\sum_{p \in P} I(a_p > 0)}{|P|}, \quad (9)$$

where $I(\cdot)$ is an indicator function that equals 1 if the condition holds and 0 otherwise.

- **Conversion Gain.** As shown in § III-B3, we aim to first enrich seed addresses in few-seed BGP prefixes and then other TGAs use enriched active addresses for target generation, thereby improving their accuracy. *Conversion Gain (CG)* measures the improvement in overall *prefix hit rate* achieved by this two-stage strategy in the few-seed BGP prefixes, compared with applying TGAs alone. Figure 5 shows the entire process. A preliminary module *pre* (e.g., 6Vision) first spends a fraction $p\%$ of the probing budget, and the remaining fraction $(100 - p)\%$ is used by a TGA τ (e.g., 6Scan) in the second stage. The active addresses discovered in stage one, together with the initial seed addresses, are used as input to stage two. Let $\text{PHR}_j^{(i)}$ denote the *prefix hit rate* achieved by algorithm j under method i (method 1: τ only; method 2: two-stage). The *CG* is defined as:

$$CG = \frac{p\% \cdot \text{PHR}_{pre}^{(2)} + (100 - p)\% \cdot \text{PHR}_\tau^{(2)}}{\text{PHR}_\tau^{(1)}} - 1. \quad (10)$$

3) *Environment:* All experiments were conducted on a cloud server equipped with $2 \times$ RTX 5090 GPUs (32 GB), 50 Intel Xeon Platinum 8470Q vCPUs, and 240 GB RAM.

B. Parameter Selection

1) *Number of Clusters:* To select the optimal number of clusters, we use the Silhouette Coefficient [60] as the quantitative evaluation metric. The Silhouette Coefficient jointly captures intra-cluster cohesion and inter-cluster separation, ranging from -1 to 1 , where values closer to 1 indicate a more well-formed clustering structure. As shown in Table III, in the

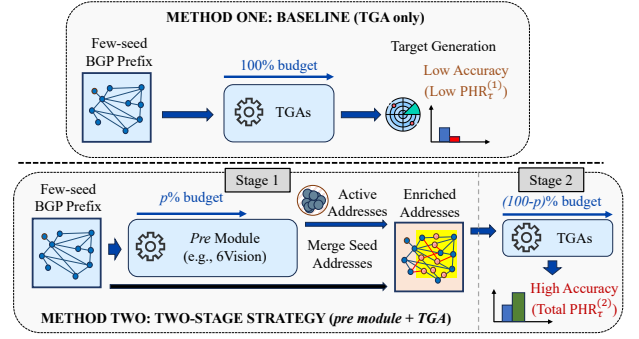


Fig. 5. The specific definition of Conversion Gain (CG).

TABLE III
SILHOUETTE COEFFICIENT FOR DIFFERENT NUMBERS OF CLUSTERS

k	3	4	5	6	7	8	9
Sil.	0.0613	0.0590	0.0620	0.0652	0.0639	0.0625	0.0600

few-seed BGP prefixes, $k = 6$ (Sil.=0.0652) provides the best trade-off between cluster compactness and cluster separability. Therefore, all subsequent experiments adopt six clusters.

2) *Parameter selection in feature augmentation:* As described in § IV-C, we encode multiple seed addresses from the same cluster into a $n \times 128$ image representation, where n denotes the number of concatenated addresses. The choice of n is critical to the performance of the proposed method. If n is too large, it may introduce redundant information, which can slightly degrade performance. In contrast, if k is too small, the constructed image contains insufficient cross-address contextual information, which weakens the benefit of feature augmentation and limits the model's ability to capture shared structural patterns among similar addresses. To determine an appropriate value of n , we conduct a parameter study by varying n over $\{2, 4, 6, 8, 10\}$. For each setting, we evaluate the performance of the model using both *hit rate* and *prefix hit rate*. In this experiment, we configure 6Vision with a probing budget of 400K and perform a single round of reinforcement learning feedback after 200K addresses have been generated. The results are shown in Figure 6. Both metrics improve consistently as n increases from 2 to 6, indicating that incorporating more addresses within each augmented feature helps the model better capture cross-address configuration patterns. When n is further increased beyond 6, both metrics exhibit a slight decline. Specifically, the *hit rate* increases from 7.59% at $n = 2$ to 7.97% at $n = 6$, while the *prefix hit rate* increases from 7.33% to 7.75%. However, when n is increased to 8 and 10, the *hit rate* slightly decreases to 7.96% and 7.95%, respectively, while the *prefix hit rate* drops to 7.73% for both settings. This suggests that excessively large n values provide limited additional contextual benefit and may instead introduce redundant information. Therefore, we select $n = 6$ as the default setting, since it achieves the best performance on both metrics and delivers the best effectiveness under the few-seed BGP prefixes.

3) *Other parameters:* For the remaining hyperparameters, we follow common deep-learning practice. For each subcategory, we use a VAE to extract image features with a 1×3

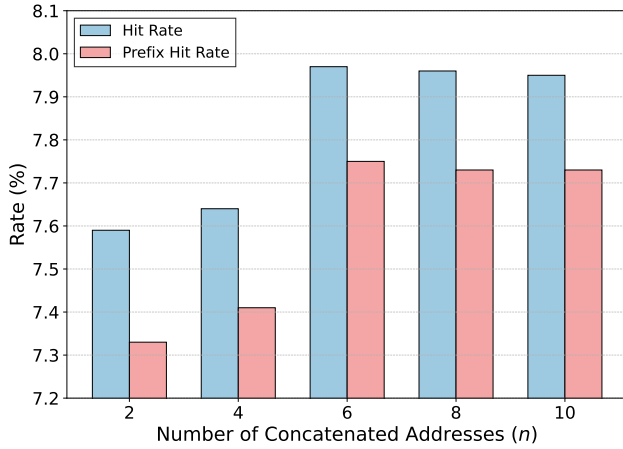


Fig. 6. Impact of the feature augmentation parameter k on hit rate and prefix hit rate.

convolution kernel and train for 200 epochs. Then we expand each cluster to three times its original size. For training the Gated PixelCNN, we use a 3×16 convolution kernel and train for 30 epochs to ensure sufficient learning. We generate image features of size 8×128 and decode the 8th row as candidate addresses. In the environment feedback mechanism, we perform one feedback round for every 200K generated addresses. The reinforcement-learning parameters are set to $\lambda_H = 0.2$ and $\lambda_{KL} = 0.1$, and we train for 5 epochs per iteration.

C. Experimental Results

1) *Comparative experiment*: We compare 6Vision with representative prior methods, including eight statistical methods, namely DET, 6Tree, 6Forest, 6Loda, AddrMiner-S, 6Scan, 6Hit, and 6Gen, as well as two machine-learning-based methods, 6GAN and 6Sense. For all baselines, we use the default parameter settings provided in their publicly available implementations. In addition, we also presented the results of the conference version 6Vision(C). The probing budgets are set to 200K, 400K, 600K, 800K, and 1000K. For 6Vision, reinforcement-learning-based feedback optimization is performed after every additional 200K probes. We first compare these methods in terms of *hit rate*, *prefix hit rate*, and *CR* under different probing budgets.

The results are shown in Figure 7. 6Vision achieves the best performance across all budgets and metrics. Moreover, 6Vision(C) consistently achieves near-second-best performance. The results differ slightly from those reported in the conference version due to changes in the network environment and the increased complexity of few-seed prefixes. Overall, we improve both model accuracy and the number of covered BGP prefixes, with the average *hit rate* increasing by 9.09%, the *prefix hit rate* by 9.58%, and the *CR* by 4.32%.

In terms of *hit rate*, 6Vision consistently outperforms all baselines and shows clear gains as the probing budget increases. Its hit rate rises steadily from 7.45% at 200K to 9.96% at 1000K, the highest among all methods. At 1000K, 6Vision improves the *hit rate* by 2%–849% over the baselines. The strongest competitor is AddrMiner-S, which reaches 9.79% at

1000K, but still falls short of 6Vision. Several statistical methods, such as DET and 6Tree, remain relatively stable across budgets, yet their performance quickly saturates and stays clearly below that of 6Vision at medium and large budgets. For example, at 1000K, DET and 6Tree achieve only 5.64% and 6.68%, respectively. By contrast, the machine-learning-based methods 6Sense and 6GAN show a clear decline as the probing budget increases, with *hit rates* of only 1.06% and 1.10%, respectively, at 1000K. This suggests that their learning effectiveness deteriorates severely in few-seed BGP prefixes, because the scarcity of seed addresses provides insufficient information for learning robust and generalizable address features. Similarly, 6Forest and 6Loda are negatively affected by their outlier-removal mechanisms, which classify about 32% of seed addresses as outliers. This removes a substantial amount of useful structural information from the original seeds and further limits their ability to learn the pattern in few-seed BGP prefixes. Overall, these results indicate that 6Vision can continuously exploit probing feedback and progressively focus on more promising regions, thereby sustaining a strong advantage in overall probing effectiveness.

A similar trend is observed for *prefix hit rate*, which more directly reflects the ability to discover active addresses within few-seed BGP prefixes. 6Vision again achieves the best performance under all probing budgets, increasing from 7.32% at 200K to 9.59% at 1000K. At 1000K, 6Vision improves the *prefix hit rate* by 45%–1,129% over the baselines. Compared with *hit rate*, its *prefix hit rate* remains very close, with a gap of only 0.13% at 200K and 0.37% at 1000K. This small gap indicates that most of the active addresses discovered by 6Vision are concentrated in the few-seed BGP prefixes, which is exactly the target scenario considered in this paper. By contrast, DET, 6Loda, 6Gen, and AddrMiner-S exhibit much larger gaps between *hit rate* and *prefix hit rate*, among which AddrMiner-S is the most pronounced. Although AddrMiner-S obtains a relatively high overall hit rate at large probing budgets, its prefix hit rate remains much lower, increasing only from 2.83% to 4.80%. This large discrepancy shows that a substantial portion of its discovered active addresses lies outside the few-seed BGP prefixes, and thus its capability for prefix-oriented enrichment is limited. Methods such as 6Tree and 6Scan show relatively competitive prefix hit rates at small or medium budgets, but their improvement is marginal compared with 6Vision. In particular, at 1000K, 6Tree and 6Scan reach only 6.62% and 6.20%, respectively. These results demonstrate that 6Vision is more effective at targeting the challenging few-seed prefixes rather than merely achieving a high overall *hit rate*.

For *CR*, which more directly reflects the ability to discover new active addresses across a broader set of few-seed BGP prefixes. 6Vision again achieves the best performance under all probing budgets, with *CR* increasing steadily from 16.24% at 200K to 26.36% at 1000K. At 1000K, 6Vision improves *CR* by 7%–278% over the baselines. The strongest competitor is 6Tree, which reaches 24.73% at 1000K, but still falls short of 6Vision. DET and 6Loda also achieve relatively high *CR* values, reaching 22.83% and 20.94%, respectively, at 1000K, yet both remain clearly below 6Vision. By contrast,

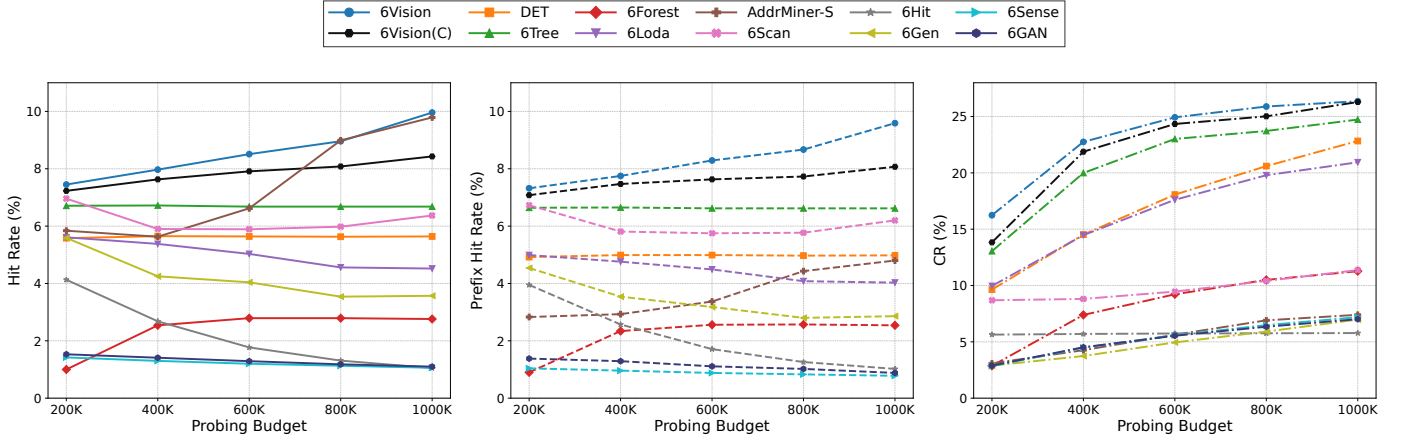


Fig. 7. Performance comparison of different methods under different probing budgets: (a) hit rate, (b) prefix hit rate, and (c) CR.

AddrMiner-S, 6Sense, 6GAN, 6Gen, and 6Hit perform much worse, with CR values of only 7.40%, 7.19%, 7.01%, 6.98%, and 5.78%, respectively, at 1000K. These low values indicate that their discovered addresses are only in a small fraction of few-seed BGP prefixes.

2) *Case study*: To better understand why 6Vision achieves excellent performance in the few-seed BGP prefixes, we conduct a case study. Using the prefixes shown in Figure 1 as examples, Figure 8 presents the corresponding seed addresses together with some active addresses discovered by 6Vision. We filtered out the addresses under these prefixes from the active addresses discovered by 6Vision. For the prefix 2001:250:307c::/48, there are only three active seed addresses. Under such sparse distributions, conventional pattern-extraction methods tend to infer overly coarse-grained patterns, as the limited and heterogeneous seeds provide insufficient evidence for precise structural characterization. As a result, the inferred patterns often overgeneralize the address space and waste probing resources on largely inactive regions. In contrast, the feature augmentation and image representation mechanism of 6Vision can capture fine-grained configuration regularities and localize active addresses more accurately. As shown in Figure 8, 6Vision successfully discovers active addresses such as 2001:250:307c:1001::1c, 2001:250:307c:1001::1, and 2001:250:307c:10::20. For the prefix 2804:14d:5000::/40, there are also only three active seeds, including 2804:14d:5086::1e. In this case, 6Vision further benefits from the reinforcement-learning-based feedback mechanism, which progressively adjusts the probing policy according to online probing results and mitigates sampling bias caused by limited initial seeds. As a result, 6Vision can quickly concentrate probing on a denser active region around the prefix and discover multiple active addresses there, such as 2804:14d:5086::185, 2804:14d:5086::be, 2804:14d:5086::7, and 2804:14d:5086::2. These examples illustrate that 6Vision improves address discovery in few-seed BGP prefixes in two complementary ways: its fine-grained image-based modeling captures structural regularities more precisely, enabling more accurate localization of active addresses and reducing wasted probing over largely inactive regions, while its online feedback optimization further mitigates sampling bias and guides

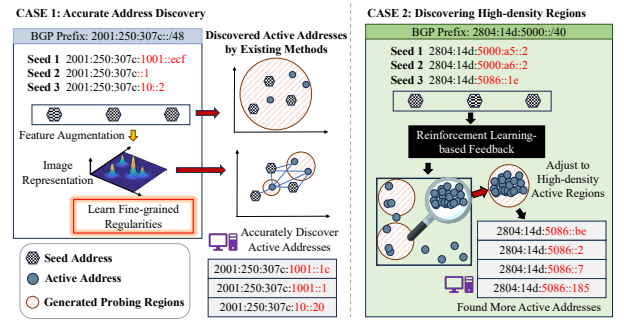


Fig. 8. Case study of active addresses discovered by 6Vision.

probing toward high-density active regions.

3) *Mechanism analysis of 6Vision*: Merely examining the final generated addresses is insufficient to reveal whether 6Vision has truly learned structural patterns within and across IPv6 addresses. To provide an intuitive explanation of the generation mechanism, we analyze how the prediction of a target bit depends on other bit positions and contextual addresses in the augmented input. Under our encoding mechanism, each sample is represented as an $n \times 128$ binary image, where each row corresponds to one IPv6 address from the same cluster and each column corresponds to a bit position. Since 6Vision models this representation autoregressively, the prediction of a target bit can only depend on its valid upper-left context under the predefined ordering. As a result, the model may exploit both *intra-row* dependencies, i.e., sequential regularities among bit positions within the same address, and *inter-row* dependencies, i.e., shared structural patterns across related addresses. To capture such dependencies, we adopt *Integrated Gradients* (IG) [61] to quantify the contribution of each input bit to the prediction of a target bit. Let $F(\mathbf{x})$ denote the model output for the target bit, where $\mathbf{x} \in \mathbb{R}^{n \times 128}$ is an input sample and $\mathbf{x}' \in \mathbb{R}^{n \times 128}$ is a baseline input. For an input position (r, c) , where r indexes the address row and c indexes the bit column, the attribution is defined as

$$IG_{r,c}(\mathbf{x}) = (x_{r,c} - x'_{r,c}) \int_0^1 \frac{\partial F(\mathbf{x}' + \alpha(\mathbf{x} - \mathbf{x}'))}{\partial x_{r,c}} d\alpha. \quad (11)$$

Unlike ordinary gradients, which measure only the model's sensitivity at the input itself, IG accumulates gradients along

the interpolation path from the baseline $\mathbf{x}'$ to the actual input $\mathbf{x}$, yielding more stable attributions. In our analysis, we use $\mathbf{x}' = \mathbf{0.5}$ as the baseline, i.e., $x'_{r,c} = 0.5$ for all r and c . This choice provides a neutral reference, since in binary IPv6 representations both 0 and 1 carry structural information, and thus neither can be regarded as a no-information state. Arranging $\text{IG}_{r,c}(\mathbf{x})$ over all rows and columns yields a two-dimensional attribution map, in which rows correspond to addresses in the input set, columns correspond to bit positions, and each value indicates the contribution of the corresponding contextual bit to the prediction of the target bit. Larger values indicate stronger influence.

Figure 9 illustrates the attribution maps for two representative intermediate generation steps when synthesizing the address 2804:164:e500::5 under the BGP prefix 2804:164:e500::/40, namely target bits 113 and 41. The results provide evidence for the two types of dependencies that our model is designed to capture, i.e., intra-row dependency within an address and inter-row dependency across addresses in the same augmented feature. From the intra-row perspective, the attribution is strongly concentrated on the already generated prefix of the target row. For target bit 113, which lies in the IID part of the address, the dominant contribution comes from the generated row itself, especially from bits 58–112. This indicates that the prediction of the current bit mainly depends on its previously generated neighboring region, consistent with the autoregressive assumption that later bits are conditioned on earlier bits in the same address. Similarly, for target bit 41, the strongest intra-row attribution is observed on bits 0–40 of the generated row, showing that the model relies heavily on the already generated prefix when predicting an earlier-position bit. From the inter-row perspective, the attribution maps also show substantial contributions from the preceding seed addresses in the augmented feature. For target bit 113, the upper rows mainly contribute through their higher-order region, roughly bits 58–128. For target bit 41, the influential range from previous rows spans approximately bits 0–98. This suggests that, beyond the target row itself, the model also exploits structurally related context from other addresses in the same cluster. Such a pattern is highly consistent with our feature augmentation strategy: by concatenating similar addresses row-wise into image representations, the model is exposed to shared configuration regularities that are difficult to identify from isolated addresses. Overall, the attribution maps confirm that the model learns to combine local sequential context within the target address and shared structural cues across clustered addresses. This observation aligns well with the design rationale of our method: feature augmentation enriches the representation by organizing similar addresses into a joint image-like input, while Gated PixelCNN captures both the intra-row autoregressive dependency and the inter-row contextual dependency through masked convolutions and an expanding receptive field. Therefore, the model is able to exploit fine-grained configuration rules for target generation.

4) *Conversion gain evaluation*: In addition to achieving high accuracy, another important advantage of 6Vision is its ability to convert few-seed BGP prefixes by expanding the available seed addresses. Such conversion is valuable because

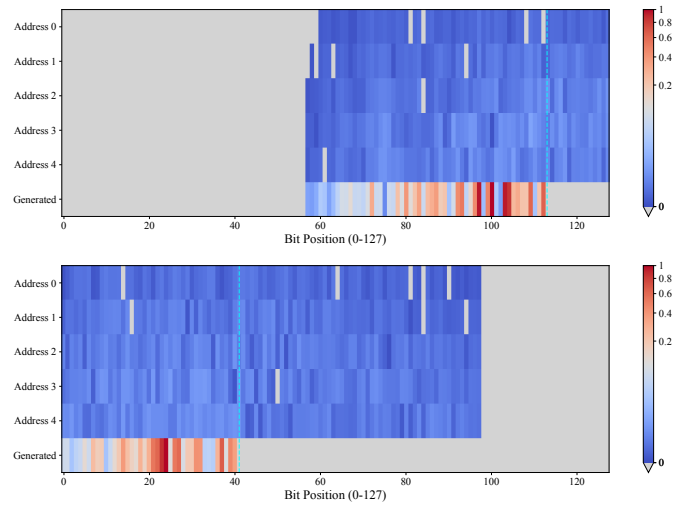


Fig. 9. Attribution maps for two examples: bit 113 (top) and bit 41 (bottom).

it can provide more informative input to existing TGAs, thereby improving their probing performance. As shown in § V-C1, among existing TGAs, 6Tree and 6Scan achieve the best performance on few-seed BGP prefixes. We therefore select these two methods to evaluate the *CG* brought by 6Vision. Specifically, we adopt a two-stage framework. In the first stage, a preliminary module consumes $p\%$ of the total probing budget to expand addresses within the few-seed BGP prefixes. In the second stage, a TGA uses the remaining $(100 - p)\%$ budget for subsequent address generation and probing. We set $p \in \{5, 10, 15, 20, 25\}$. The total probing budget is set to 4000k. When the entire budget is directly used by 6Tree and 6Scan, their performance is poor, with *prefix hit rates* of only 3.95% and 5.49%, respectively. This suggests that existing methods struggle to learn fine-grained address-configuration patterns from the initial seed addresses. In contrast, the proposed two-stage strategy first enriches the seed addresses and then allows the downstream TGA to exploit the improved prefix representation.

Table IV reports the results under different parameter settings. For each configuration, the number of Active Addresses is divided into two parts: the first part denotes the active addresses discovered by the preliminary module using $p\%$ of the budget, and the second part denotes those further discovered by the downstream TGA using the remaining $(100 - p)\%$ budget. Their sum is the total number of active addresses discovered. *CG* measures the relative improvement of the two-stage strategy over directly applying the corresponding TGA. The results show that when 6Vision is used as the preliminary module, it consistently brings substantial positive *CG* for downstream TGAs. In particular, the combination of 6Vision and 6Tree achieves the greatest improvement across all tested p values, with *CG* ranging from 241.52% to 330.38%, and reaching the maximum at $p = 10$. This indicates that 6Vision can effectively enrich these prefixes with high-quality addresses, enabling 6Tree to learn more informative structural patterns in the second stage. For 6Vision+6Scan, although the *CG* is smaller than that of 6Vision+6Tree, the improvement remains consistently positive, with *CG* ranging

TABLE IV
ACTIVE ADDRESSES (K) AND CONVERSION GAIN (CG) UNDER DIFFERENT PARAMETER SETTINGS

Pre module	TGAs	Metric	$p = 5$	$p = 10$	$p = 15$	$p = 20$	$p = 25$
6Vision	6Tree	Active Addresses (k)	14.64 + 609.54	31.00 + 649.06	49.74 + 621.30	69.36 + 552.27	95.90 + 443.68
		CG	294.94%	330.38%	324.81%	293.42%	241.52%
	6Scan	Active Addresses (k)	14.64 + 355.25	31.00 + 465.50	49.74 + 412.04	69.36 + 347.17	95.90 + 273.30
		CG	68.49%	126.05%	110.18%	89.62%	68.12%
6Tree	6Tree	Active Addresses (k)	13.40 + 135.53	26.60 + 115.13	39.72 + 113.08	52.96 + 102.87	66.20 + 95.42
		CG	-5.82%	-10.38%	-3.29%	-1.27%	2.28%
	6Scan	Active Addresses (k)	13.40 + 199.80	26.60 + 184.43	39.72 + 174.65	52.96 + 159.36	66.20 + 149.73
		CG	-2.91%	-3.83%	-2.37%	-3.28%	-1.64%
6Scan	6Tree	Active Addresses (k)	13.46 + 121.41	23.24 + 114.03	34.50 + 111.55	46.16 + 102.77	63.70 + 107.68
		CG	-14.68%	-13.16%	-7.59%	-5.82%	8.35%
	6Scan	Active Addresses (k)	13.46 + 197.62	23.24 + 183.68	34.50 + 180.89	46.16 + 170.35	63.70 + 154.51
		CG	-3.83%	-5.83%	-2.00%	-1.46%	-0.55%

from 68.12% to 126.05%. This demonstrates that 6Vision serves as an effective enhancement module for different downstream TGAs. By contrast, when 6Tree or 6Scan is used as the preliminary module, the *CG* is mostly negative or close to zero. This suggests that existing TGAs are not well-suited for the conversion of few-seed BGP prefixes. While they can discover some active addresses, they tend to learn only coarse-grained patterns, making them less capable of uncovering addresses in dense regions and therefore less effective at enhancing the performance of downstream TGAs. We also observe that the gain brought by 6Vision is more pronounced when only a small fraction of the budget is allocated to the preliminary stage, especially at $p = 10$ or $p = 15$, where the best or near-best results are obtained. This indicates that only a modest budget is needed for 6Vision to achieve effective gains. Overall, these results demonstrate that 6Vision is not only effective as a standalone TGA in few-seed BGP prefixes, but also serves as a general and practical preliminary module that significantly improves the performance of existing TGAs on few-seed BGP prefixes.

5) *Effectiveness of clustering methods*: To rigorously evaluate the effectiveness of our clustering algorithm and compare it with alternative clustering methods, we adopt the Multi-Resolution Aggregate (MRA) count ratio proposed by Plonka *et al.* [62]. This metric is particularly suitable for IPv6 address analysis because the structural regularity of IPv6 addresses is primarily manifested in how active addresses concentrate under prefixes of different lengths. The MRA count ratio captures this hierarchical aggregation behavior directly across prefix levels, unlike geometric distance metrics such as Euclidean distance.

Let n_p denote the number of active aggregates at prefix length $/p$, i.e., the number of distinct $/p$ prefixes covered by an address set. The single-bit MRA count ratio is defined as

$$\gamma_p = \frac{n_{p+1}}{n_p}, \quad p = 0, 1, \dots, 127, \quad (12)$$

where $1 \leq \gamma_p \leq 2$. Intuitively, a γ_p value close to 1 indicates that increasing the prefix length by one bit introduces few new prefixes, suggesting a denser concentration of addresses at that bit position. In contrast, a value close to 2 indicates a more dispersed distribution.

To summarize the dispersion of multiple address sets, we compute the normalized area under the MRA count ratio curve for each cluster. For the i -th cluster, we define

$$D_i = \frac{1}{128} \sum_{p=0}^{127} (\gamma_p^{(i)} - 1), \quad (13)$$

where $D_i \in [0, 1]$, and a larger value indicates higher within-cluster dispersion. We then compute the overall size-weighted mean dispersion across all clusters as

$$\bar{D} = \frac{\sum_i |S_i| D_i}{\sum_i |S_i|}, \quad (14)$$

where $|S_i|$ denotes the number of addresses in cluster S_i . A smaller $\bar{D}$ indicates that addresses are more tightly grouped within clusters, implying better clustering quality.

We compare our method with several representative clustering algorithms. Specifically, we include k-means, which is used in 6GAN; DHC, which is adopted by methods such as 6Tree and 6Scan; AHC, which is used in 6Gen; and GMM, a widely used general-purpose clustering method. We compute the corresponding $\bar{D}$ values for all these methods, as shown in Table V. Compared with GMM, our method reduces these fluctuations across a wider range of prefix lengths, yielding consistently smaller γ_p values and therefore more compact clusters under the MRA-based criterion. Among all compared methods, our clustering method achieves the best overall performance, with the lowest size-weighted mean dispersion $\bar{D}$ of 0.0985, indicating that it produces clusters that are more compact. To provide an intuitive comparison, we plot the MRA count ratio of our method and GMM as representative examples in Figure 10, where n represents the number of addresses in the cluster. Overall, our method exhibits curves that stay closer to 1, so hence lower within-cluster dispersion. It is also worth noting that, for both methods, γ_p fluctuates substantially at shorter prefix lengths (before $/48$), indicating the diversity of prefixes. Beyond approximately $/48$, the curves largely stabilize near 1, suggesting that many clusters share consistent $/48$ -level aggregation. Notably, we still observe more visible fluctuations at longer prefix lengths (around $/112$ – $/128$), which is consistent with variability in the lower bits (e.g., interface identifier construction), and thus naturally leads to higher dispersion in that range.

TABLE V
COMPARISON OF CLUSTERING METHODS USING SCORE $\bar{D}$.

Method	Ours	DHC	AHC	GMM	k-means
$\bar{D}$	0.0985	0.1041	0.1047	0.1006	0.0986

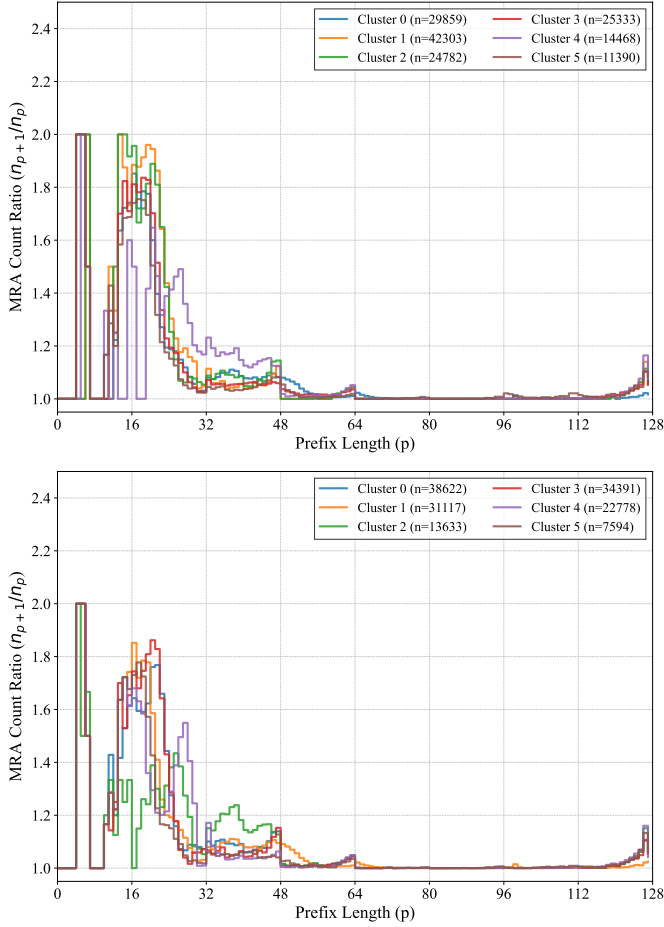


Fig. 10. MRA count-ratio curves for GMM (top) and our method (bottom).

6) *The impact of different encoding granularities:* 6Vision learns from IPv6 addresses in their 128-bit binary form. To examine the effect of encoding granularity on model performance, we further consider two commonly used alternatives [26], namely, the 64-digit quaternary representation and the 32-digit hexadecimal representation. Concretely, we modify the address formulations in § IV-C and § IV-D by replacing the original 128-bit binary form with the corresponding quaternary or hexadecimal form, and update the value range of each position accordingly. We evaluate the three variants using *hit rate*, *prefix hit rate*, and *CR*.

Table VI shows that different encoding granularities exhibit distinct behaviors under different probing budgets. The binary representation performs particularly well under small budgets. At 200K, it achieves the best results on all three metrics, with 7.45% *hit rate*, 7.32% *prefix hit rate*, and 16.24% *CR*. The quaternary representation becomes increasingly competitive as the probing budget grows. It achieves the highest *hit rate* at 400K, 600K, and 1000K, and also attains the best *prefix hit rate* at multiple budget settings. This suggests that quater-

nary encoding provides a favorable intermediate granularity: compared with the binary representation, it reduces sequence length and may ease pattern learning, while still preserving sufficient structural information to support accurate generation. The hexadecimal representation also shows strengths, especially at 800K, where it obtains the highest overall *hit rate*. However, its *prefix hit rate* and *CR* are generally lower than those of the other two representations, suggesting that this coarser granularity is less effective at exploring few-seed BGP prefixes and cover fewer. From the perspective of prefix-level coverage, the binary representation remains the most excellent. It achieves the highest *CR* across all budget settings, increasing from 16.24% at 200K to 26.36% at 1000K. Since a larger *CR* indicates that active addresses are discovered in more few-seed BGP prefixes, this result suggests that the binary representation is better aligned with objective of expanding coverage across few-seed prefixes. In contrast, while the quaternary and hexadecimal representations can be competitive in terms of *hit rate* under some budgets, they are less effective at covering a broader set of few-seed BGP prefixes. In addition, for the quaternary and hexadecimal representations, the gap between *hit rate* and *prefix hit rate* is generally larger than that of the binary representation. This indicates that these encodings are more likely to generate active addresses outside the few-seed BGP prefixes, rather than finely concentrating on the target BGP prefix space. As a result, they are less effective at maintaining probing focus on few-seed BGP prefixes, which further limits their ability to support prefix coverage.

D. Ablation Study

The core design of 6Vision includes two components to address the two major challenges in few-seed BGP prefixes, namely, sparse seed-address distribution and severe sampling bias. The first is *feature augmentation*, which concatenates similar addresses into image representations so that the model can learn fine-grained dependencies both within and across addresses. This design enriches learnable features and improves the model's ability to learn effective representations under sparse seed distribution. The second is *feedback optimization*, which introduces reinforcement learning to adapt the generation policy according to probing feedback, enabling the model to progressively discover high-density regions during probing and thereby mitigating the impact of severe sampling bias in the initial seeds.

To assess the contribution of these two components, we conduct an ablation study by removing feature augmentation, feedback optimization, or both, and then compare the resulting variants with the full 6Vision model in terms of *hit rate*, *prefix hit rate*, and *CR* under different probing budgets. The results are in Figure 11. First, removing feature augmentation causes a substantial performance drop. For example, at a probing budget of 1000K, the *hit rate* decreases from 9.96% to 9.13%, the *prefix hit rate* decreases from 9.59% to 7.29%, and the *CR* drops sharply from 26.36% to 18.05%. Similar gaps are observed across the entire budget range. These results indicate that feature augmentation is critical for extracting

TABLE VI
IMPACT OF DIFFERENT ENCODING GRANULARITIES UNDER DIFFERENT PROBING BUDGETS.

Budget	Hit Rate (%)			Prefix Hit Rate (%)			CR (%)		
	Bin.	Quat.	Hex.	Bin.	Quat.	Hex.	Bin.	Quat.	Hex.
200K	7.45	6.65	4.64	7.32	6.26	3.94	16.24	13.73	8.97
400K	7.97	9.42	7.95	7.75	8.73	6.64	22.75	19.05	13.34
600K	8.51	9.65	8.99	8.29	8.75	7.29	24.93	21.45	15.26
800K	8.96	9.94	11.64	8.67	10.97	8.95	25.89	22.45	16.10
1000K	9.96	14.68	11.79	9.59	12.62	8.85	26.36	22.76	16.59

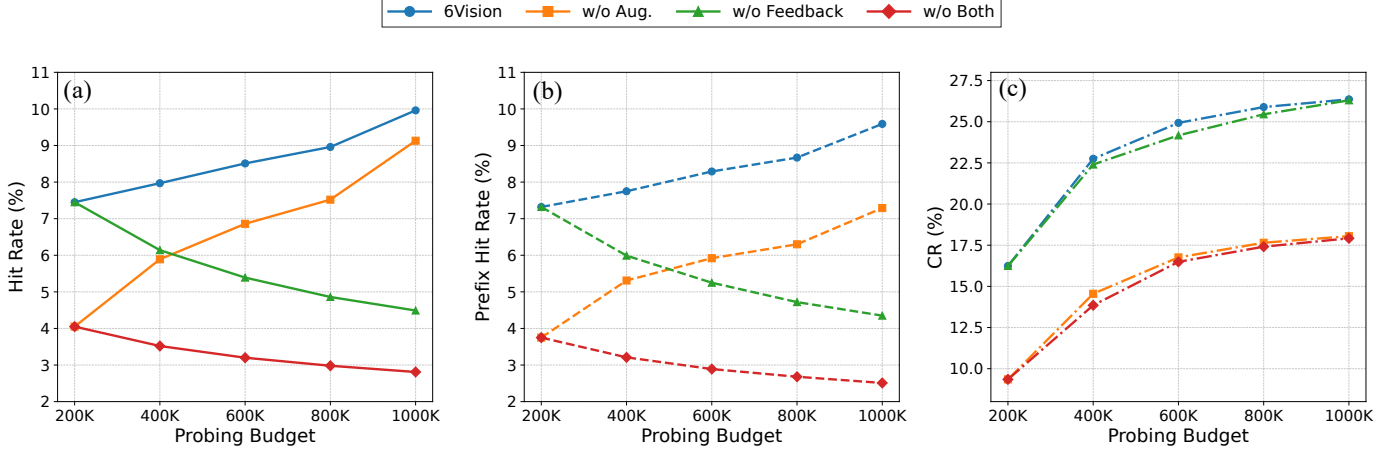


Fig. 11. Ablation study of 6Vision under different probing budgets: (a) hit rate, (b) prefix hit rate, and (c) CR.

discriminative features from sparse seed addresses. In the absence of this component, weakened representation learning degrades all three metrics. The most pronounced reduction is observed in *CR*, implying that the model loses diversity in generated target addresses. In addition, the gap between *hit rate* and *prefix hit rate* becomes noticeably larger. At 1000K, this gap reaches 1.83%, suggesting that without feature augmentation, the generated active addresses are less effectively concentrated within few-seed BGP prefixes. Moreover, feedback optimization continuously improves performance, especially as the probing budget increases. Compared with the full model, removing feedback optimization leads to a persistent decline in both hit rate and prefix hit rate. At 1000K, the hit rate decreases from 9.96% to 4.49%, the prefix hit rate decreases from 9.59% to 4.35%, while the *CR* only slightly decreases from 26.36% to 26.30%. These results suggest that feedback optimization is essential for progressively adapting the generation policy according to online probing results, allowing the model to focus more effectively on dense and promising regions. Its benefit is particularly evident in hit rate and prefix hit rate, where dynamic adjustment can continuously correct the bias introduced by the initial seeds. When both components are removed, the performance becomes the worst in almost all cases. At 1000K, the hit rate and prefix hit rate fall to 2.81% and 2.51%, respectively, while the *CR* is only 17.92%. Feature augmentation mainly strengthens representation learning and helps preserve the diversity of discovered active addresses, thereby discover addresses under more few-seed BGP prefixes, whereas feedback optimization improves continuous exploration during probing. Their combi-

nation enables 6Vision to maintain superior effectiveness and robustness across different probing budgets.

E. Large-Scale Deployment

To comprehensively evaluate the Internet-scale probing capability of 6Vision and enrich the address space of few-seed BGP prefixes, we continuously deployed 6Vision for large-scale probing from November 2025 to April 2026. We constructed a hitlist by merging all active addresses discovered by 6Vision during this period. As shown in Table VII, the deployment discovers 33.08M non-aliased active IPv6 addresses in few-seed BGP prefixes. Specifically, the top five BGP prefixes account for 44.18% of all discovered addresses, while the top five ASes account for 89.58%. Cloudflare and Amazon dominate the deployment results, indicating that large providers contribute a substantial portion of the active address space in few-seed BGP prefixes. Table VIII further shows that the discovered addresses span multiple configuration patterns identified by *addr6* [63]. Embedded-ipv4 and randomized patterns are the most prevalent, accounting for 33.02% and 31.13% of the hitlist, respectively. We note that the randomized pattern identified here follows *addr6*-style pattern classification and is therefore not strictly equivalent to uniformly random generation; rather, it refers to addresses that do not exhibit stable low-byte, embedded, or other explicit structural regularities. Importantly, all reported addresses have been strictly filtered by APD, meaning that the randomized category is derived exclusively from non-aliased active addresses. This finding suggests that a considerable number of active hosts in few-seed BGP prefixes still adopt entropy-rich or privacy-

oriented interface identifier generation. Low-byte and pattern-bytes account for 15.41% and 15.13%, respectively, whereas ieee-derived and embedded-port patterns are much less common. These results highlight the structural diversity of active IPv6 addresses in few-seed BGP prefixes, helping explain why existing coarse-grained pattern learning methods often achieve limited accuracy and underscoring the necessity of fine-grained modeling for effective large-scale discovery. Our collected hitlist is valuable as it pushes the coverage boundary of existing IPv6 hitlists toward previously hard-to-explore few-seed BGP prefixes, thereby supporting broader future research on IPv6 address space structure and network behavior.

TABLE VII
SUMMARY OF LARGE-SCALE DEPLOYMENT RESULTS.
TOP ROW: BGP PREFIXES; BOTTOM ROW: ASes.

#Addrs	TOP1	TOP2	TOP3	TOP4	TOP5
33.08M	23.57% 	6.64% 	6.27% 	3.90% 	3.59% 
	50.08% 	26.08% 	7.17% 	3.38% 	2.87% 

 Cloudflare  Amazon  Akamai  HAProxy

TABLE VIII
DISTRIBUTION OF ADDRESS CONFIGURATION PATTERNS

Pattern	Percentage
embedded-ipv4	33.02%
randomized	31.13%
low-byte	15.41%
pattern-bytes	15.13%
ieee-derived	5.11%
embedded-port	0.20%

VI. ETHICAL CONSIDERATIONS

We adhere to the ethical principles outlined in the Belmont Report [64] and the Menlo Report [65], which emphasize respect for persons, beneficence, justice, and respect for law and public interest. We also draw on the ethical scanning practices established by ZMap [3], 6Scan [17], AddrMiner [16], and 6Sense [9]. Specifically, we carefully limit the scan rate to below 10k pps and randomly select targets to minimize potential interference. To further reduce the impact of probing on target hosts, we send only one probe to each IP address. Our measurements are limited to ICMPv6 scanning: we do not collect service banners or digital certificates, nor do we perform vulnerability probing or any form of device modification. In addition, all scanning machines provide identifiable DNS and web records containing contact information and opt-out instructions. No complaints or opt-out requests were received during our experiments.

VII. CONCLUSION

IPv6 active address discovery is important for Internet measurement, asset census, and security analysis, but existing target generation methods often perform poorly in few-seed BGP prefixes due to sparse seeds and sampling bias. In this paper, we present 6Vision, a fine-grained IPv6 active address discovery method designed for such scenarios. By clustering structurally similar seed addresses, constructing image-based augmented representations, learning dependencies both within and across addresses, and incorporating reinforcement-learning-based online feedback, 6Vision improves pattern

learning under sparse seeds and continuously refines probing toward more promising active regions. Overall, 6Vision provides an effective solution for IPv6 active address discovery in few-seed BGP prefixes and helps enrich these underexplored regions for more efficient subsequent probing. We believe that 6Vision, together with the hitlist collected in this work, provides both an effective methodological foundation and a valuable data resource for future research in network measurement, Internet-wide asset discovery, and security analysis.

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